Thermodynamics and Statistical Mechanics

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Thermodynamics

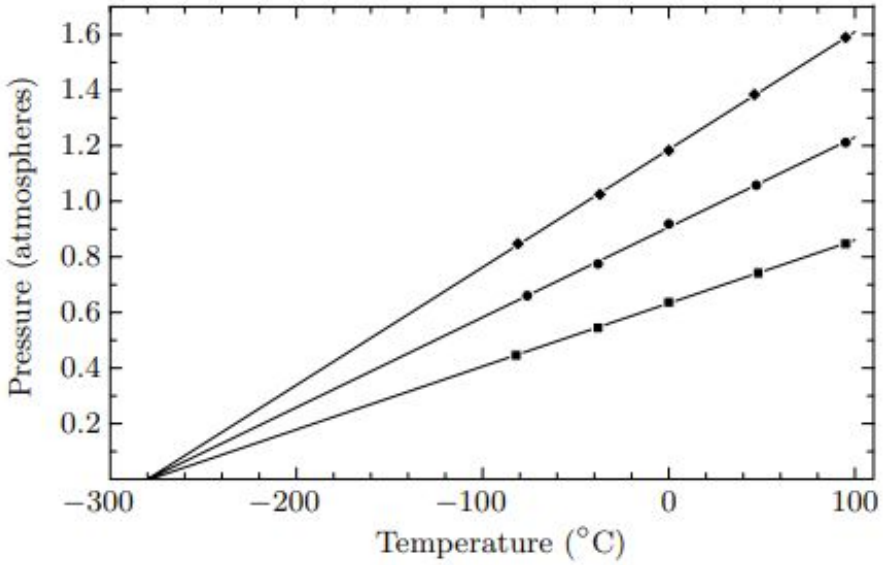
- Some of the properties of bulk matter don't really depend on the microscopic details of atomic physics
- Heat always flows spontaneously from a hot object to a cold one, never the other way
- The maximum possible efficiency of an engine, working over a given temperature range, is the same whether the engine uses steam or air or anything else as its working substance
- These kinds of results, and the principles that generalize them, comprise a subject called **thermodynamics**

Statistical Mechanics

- To understand matter in more detail, we must also take into account both the quantum behavior of atoms and the laws of statistics that make the connection between one atom and 10^{23}
- Then, we can not only predict the properties of metals and other materials, but also explain why the principles of thermodynamics are what they are
- This underlying explanation of thermodynamics comprise a subject called **statistical mechanics**

Temperature and Thermal equilibrium

- Temperature is measure of tendency of an object to spontaneously give up energy to its surroundings
- When two objects are in thermal contact, the one that tends to spontaneously lose energy is at the higher temperature.
- When two objects are in contact for long enough, they are in thermal equilibrium
- *Contact* here means some ways for two objects to exchange energy spontaneously, in the form we call “heat”
- It isn’t necessary to have exchange of particles to reach thermal equilibrium (this is needed for chemical or diffusive equilibrium)
- Absolute temperature

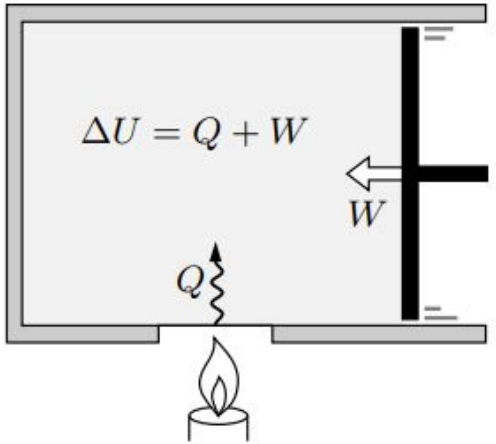


Heat and Work

- **Heat** is defined as any spontaneous flow of energy from one object to another, caused by a difference in temperature between the objects
- **Work**, in thermodynamics, is defined as any other transfer of energy into or out of a system.
- Usually, with work, we can identify some “agent” (possibly an inanimate object) that is “actively” putting energy into the system; it wouldn’t happen “automatically”
- Both heat and work refer to energy in transit

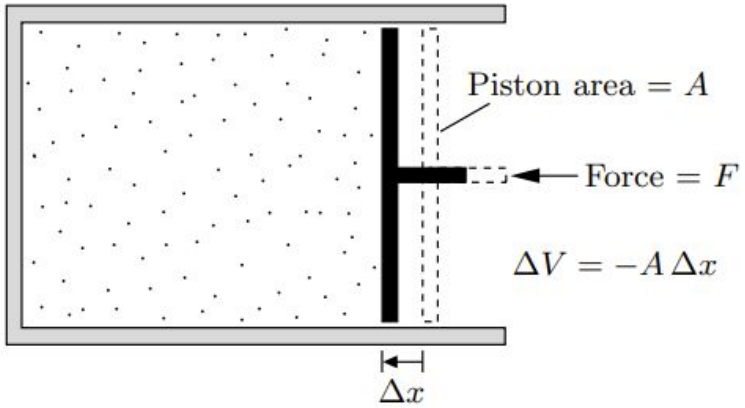
1st law of thermodynamics

- The total change in the energy (U) of a system is the sum of the heat (Q) added to it and the work done on it (W)



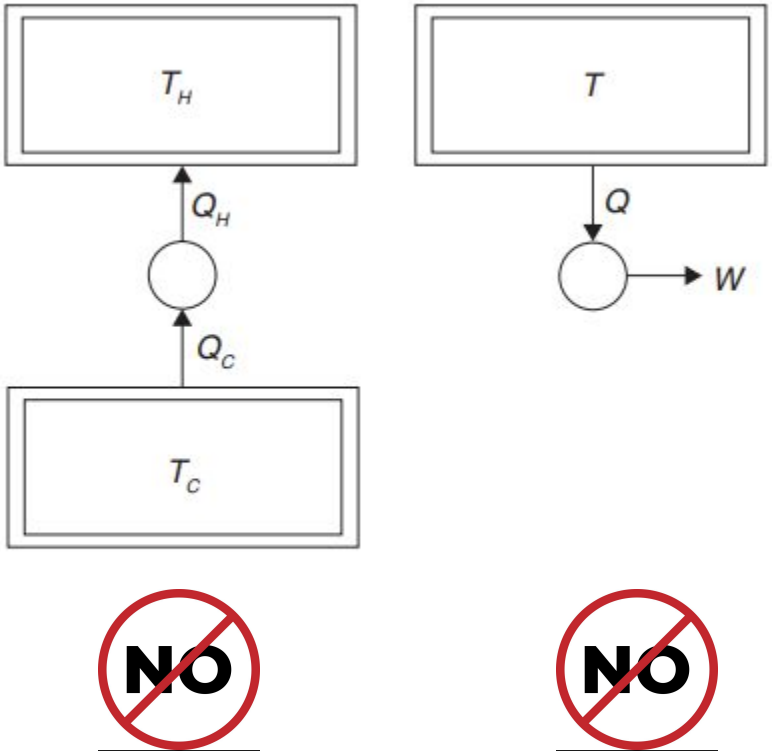
- **Q < 0** => the system is cooled rather than heated
- **W < 0** => work is done by rather than on the system

- When the piston moves inward, the volume of the gas changes by ΔV and the work done on the gas (assuming quasi-static compression) is -PΔV



- Compression-expansion work not the only type of work that can be done on thermodynamic systems (the chemical reactions in a battery cause electrical work to be done on the circuit it is connected to)

2nd law of thermodynamics



Clausius' version - A process whose only result is to cause one heat reservoir with temperature T_C to lose energy Q_C through cooling and cause a hotter heat reservoir with temperature $T_H > T_C$ to gain energy Q_H through heating is forbidden

Kelvin's version - A heat engine whose only result is to cause a single heat reservoir to lose energy Q through cooling and perform an equal amount of work W [= Q] is forbidden

Why does heat flow spontaneously from a hotter object to a cooler object, never the other way?

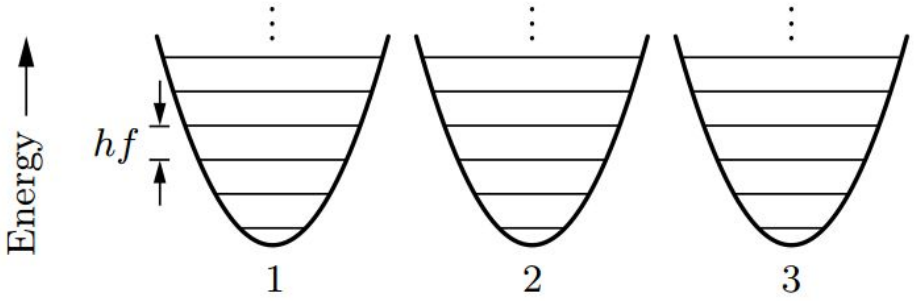
Microstates and Macrostates

- Each of eight different outcomes is called a **microstate**
- To specify the **microstate** of a system, we must specify the state of each individual particle, in this case the state of each coin
- If we specify the state more generally, by merely saying how many heads or tails there are, we call it a **macrostate**
- The number of microstates corresponding to a given macrostate is called the **multiplicity (Ω)** of that macrostate
- Ω (2 heads) = 3

Penny	Nickel	Dime
H	H	H
H	H	T
H	T	H
T	H	H
H	T	T
T	H	T
T	T	H
T	T	T

Counting microstates for an Einstein solid

- Solid is modeled as a collection of oscillators (potential wells) with evenly spaced energy levels
- **Problem:** How to distribute q units of energy among N oscillators?



Oscillator:	#1	#2	#3
Energy:	0	0	0
q=1	1	0	0
	0	1	0
	0	0	1
q=2	2	0	0
	0	2	0
	0	0	2
	1	1	0
	1	0	1
	0	1	1

Oscillator:	#1	#2	#3
Energy:	3	0	0
q=3	0	3	0
	0	0	3
	2	1	0
	2	0	1
	1	2	0
	0	2	1
	1	0	2
	0	1	2
	1	1	1

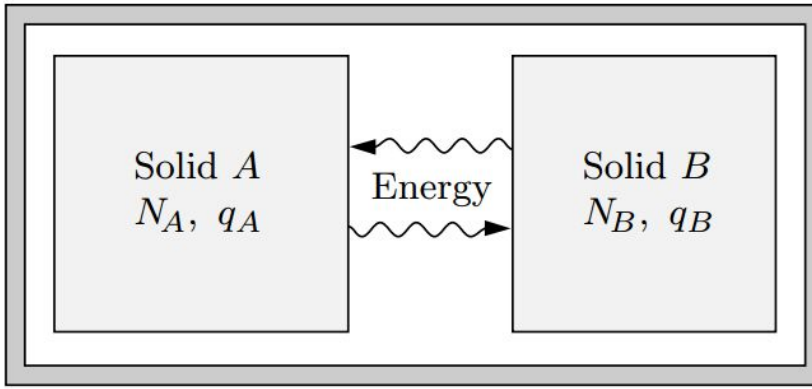
Assumption: All allowed energy levels have equal probability

Putting q indistinguishable objects in N boxes

$$\Omega(N,q) = (q+N-1)C_{N-1}$$

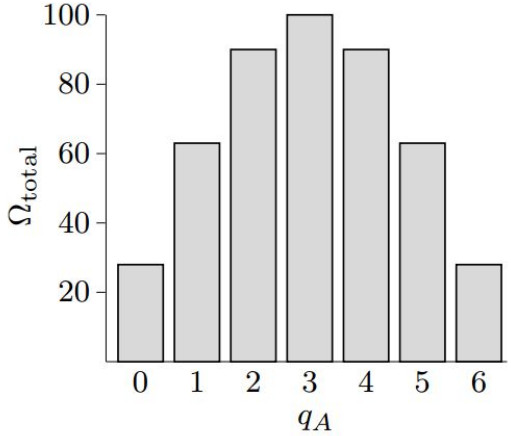
Counting multiplicity in an interacting system

- **To understand heat flow and irreversible processes**, consider a system of two Einstein solids that can exchange energy
- Consider $N_A = N_B = 3$; $q_{total} = q_A + q_B = 6$
- Total multiplicity of any macrostate, $\boldsymbol{\Omega}_{total}$, is the product of individual multiplicities, since the systems are independent of each other



In an isolated system in thermal equilibrium, all accessible microstates are equally probable. But some macrostates are more probable than others

q_A	Ω_A	q_B	Ω_B	$\Omega_{\text{total}} = \Omega_A\Omega_B$
0	1	6	28	28
1	3	5	21	63
2	6	4	15	90
3	10	3	10	100
4	15	2	6	90
5	21	1	3	63
6	28	0	1	28
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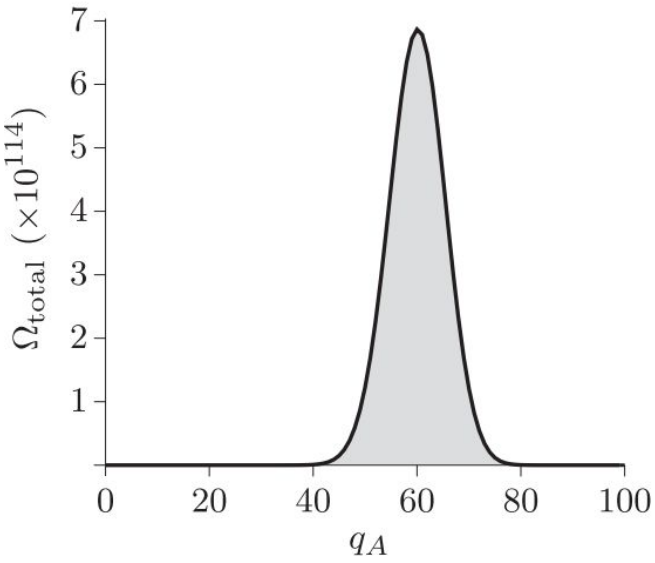


Counting multiplicity in an interacting system

N_A =300, N_B = 200

$q_{total} = q_A + q_B$ =100

q_A	Ω_A	q_B	Ω_B	Ω_{total}
0	1	100	2.8×10^{81}	2.8×10^{81}
1	300	99	9.3×10^{80}	2.8×10^{83}
2	45150	98	3.1×10^{80}	1.4×10^{85}
3	4545100	97	1.0×10^{80}	4.6×10^{86}
4	3.4×10^{8}	96	3.3×10^{79}	1.1×10^{88}
⋮	⋮	⋮	⋮	⋮
59	2.2×10^{68}	41	3.1×10^{46}	6.7×10^{114}
60	1.3×10^{69}	40	5.3×10^{45}	6.9×10^{114}
61	7.7×10^{69}	39	8.8×10^{44}	6.8×10^{114}
⋮	⋮	⋮	⋮	⋮
100	1.7×10^{96}	0	1	1.7×10^{96}
				9.3×10^{115}



- The most likely macrostate is more than 10^{33} times more probable than the least likely macrostate
- The spontaneous flow of energy stops when a system is at, or very near, its most likely macrostate, i.e. the macrostate with the greatest multiplicity

Multiplicity of a very large Einstein solid

$$\Omega(N,q) = \binom{q+N-1}{q} = \frac{(q+N-1)!}{q!\,(N-1)!} \approx \frac{(q+N)!}{q!\,N!}.$$

Using Stirling's approximation

$$N! \approx N^N e^{-N}\sqrt{2\pi N}$$

$$\ln N! \approx N\ln N - N$$

$$\ln\Omega \approx N\ln\frac{q}{N} + N + \frac{N^2}{q}$$

$$\Omega(N,q) \approx e^{N\ln(q/N)}e^N = \left(\frac{eq}{N}\right)^N \qquad (\text{when } q \gg N)$$

N	$N!$	$N^N e^{-N}\sqrt{2\pi N}$	Error	$\ln N!$	$N\ln N - N$	Error
1	1	.922	7.7%	0	−1	∞
10	3628800	3598696	.83%	15.1	13.0	13.8%
100	9×10^{157}	9×10^{157}	.083%	364	360	.89%

Table 2.3. Comparison of Stirling's approximation (equations 2.14 and 2.16) to exact values for $N = 1$, 10, and 100.

Entropy

- Since multiplicities tend to be very large numbers, entropy is defined as

  $\mathbf{S = k \ln \Omega}$, where $k = 1.38 \times 10^{-23}$ J/K

- We can restate 2nd law of thermodynamics as

  Any large system in equilibrium will be found in the macrostate with the greatest entropy

- Entropy tends to increase

- The entropy of an ideal gas depends on its energy, volume and number of particles. Increasing any of these three variables increases the entropy

  S can be written as S(U, V, N)

Condition for equilibrium

Two weakly coupled systems (A & B) are allowed to exchange quantities and reaches equilibrium where total entropy (S_A+S_B) is maximum with a constraint that total exchanged quantity is fixed

Type of interaction	**Exchanged quantity**	**Entropy maximization**	**Governing variable**	**Final condition**
Thermal	Energy	$\frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$	$\frac{1}{T} \equiv \left(\frac{\partial S}{\partial U}\right)_{N,V}$	$T_A=T_B$
Mechanical	Volume and energy	$\frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$ $\frac{\partial S_A}{\partial V_A} = \frac{\partial S_B}{\partial V_B}$	$\frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_{U,N}$	$P_A=P_B$ $T_A=T_B$
Diffusive	Particles and energy	$\frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$ $\frac{\partial S_A}{\partial N_A} = \frac{\partial S_B}{\partial N_B}$	$\frac{\mu}{T} = -\left(\frac{\partial S}{\partial N}\right)_{U,V}$	$\mu_A=\mu_B$ $T_A=T_B$

Thermodynamic identity

S= S(U, V, N)

$$dS = \left(\frac{\partial S}{\partial U}\right)_{N,V} dU + \left(\frac{\partial S}{\partial V}\right)_{N,U} dV + \left(\frac{\partial S}{\partial N}\right)_{U,V} dN$$
$$= \frac{1}{T}\,dU + \frac{P}{T}\,dV - \frac{\mu}{T}\,dN.$$

Solving for dU

$$dU = T\,dS - P\,dV + \mu\,dN$$

-PdV term is mechanical work and the μdN term is the chemical work
This combines both the 1st law and 2nd law of thermodynamics

Criteria for spontaneity (not simple)

ΔS	ΔS_{surr}	ΔS_{tot}	Character
+	+	+	spontaneous
−	−	−	not spontaneous; reverse change is spontaneous
+	−		spontaneous if ΔS dominates
−	+		spontaneous if ΔS_{surr} dominates

$$\overbrace{\Delta S_{tot}}^{\text{total entropy change}} = \overbrace{\Delta S}^{\text{entropy change of system}} + \overbrace{\Delta S_{surr}}^{\text{entropy change of surrondings}}$$

Any process is spontaneous only if the total entropy change is positive

One of the problems with using the second law of thermodynamics to judge whether a reaction is spontaneous is that to assess the total entropy change, two quantities, the entropy change of the system and the entropy change of the surroundings, must be calculated and then added together. Some of this effort could be avoided if a single property combined the entropy calculations for the system and the surroundings. This simplification is achieved by introducing a new state function, the "Gibbs free energy," probably the single most widely used and useful quantity in the application of thermodynamics to chemistry. It is named for Josiah Willard Gibbs (FIG. 4J.1), the nineteenth-century American physicist who was responsible for turning thermodynamics from an abstract theory into a subject of great usefulness.

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Gibbs Free energy

4J.1 Focusing on the System

The total entropy change, ΔS_{tot}, is the sum of the changes in the system, ΔS, and its surroundings, ΔS_{surr}, with $\Delta S_{tot} = \Delta S + \Delta S_{surr}$. For a process at constant temperature and pressure, the change in the entropy of the surroundings is given by Eq. 2 of Topic 4I ($\Delta S_{surr} = -\Delta H/T$). Therefore,

$$\Delta S_{tot} = \Delta S + \overbrace{\Delta S_{surr}}^{-\Delta H/T} = \Delta S - \frac{\Delta H}{T} \text{ at constant temperature and pressure} \quad (1)$$

This equation allows the total entropy change to be calculated from information about the system alone. The limitation is that it is valid only at constant temperature and pressure.

To take the next step, the **Gibbs free energy,** G, is introduced. It is defined as

$$G = H - TS \quad (2)$$

This quantity, which is commonly known as the *free energy* and more formally as the *Gibbs energy*, is defined solely in terms of state functions, so G itself is a state function. For a process occurring at constant temperature, the change in Gibbs free energy is

$$\Delta G = \Delta H - T\Delta S \text{ at constant temperature} \quad (3)$$

By comparing this expression rearranged into

$$\frac{\Delta G}{T} = \frac{\Delta H}{T} - \Delta S$$

with Eq. 1, where there is the additional constraint of constant pressure, we see that $\Delta G/T = -\Delta S_{tot}$ and therefore that

$$\Delta G = -T\Delta S_{tot} \text{ at constant temperature and pressure} \quad (4)$$

It is easy to predict the sign of the entropy change of the surroundings by noting whether the process is exothermic or endothermic. If the process is exothermic, then heat is released into the surroundings and their entropy increases ($\Delta S_{surr} > 0$). If the process is endothermic, heat leaves the surroundings, so their entropy decreases ($\Delta S_{surr} < 0$). To calculate the numerical value of the change, Eq. 1 of Topic 4F ($\Delta S = q_{rev}/T$) is used after noting the following points:

- Because the surroundings are always taken to be large, their temperature remains constant however much energy is transferred to or from them as heat, so the expression can always be used in the form $\Delta S_{surr} = q_{surr,\,rev}/T$.
- The heat that leaves the system enters the surroundings, $q_{surr} = -q$ (so if $q = -10$ kJ, meaning that 10 kJ has left the system, $q_{surr} = +10$ kJ, meaning that 10 kJ has entered the surroundings).
- For a system maintained at constant pressure the heat that leaves the system can be equated to the change in enthalpy of the system, so $q = \Delta H$, and therefore $q_{surr} = -\Delta H$.
- Because the surroundings are large, any transfer of heat into them will, from their perspective, be so tiny that it can be regarded as occurring reversibly, and hence $q_{surr,\,rev} = -\Delta H$.

It follows from these points that

$$\Delta S_{surr} = -\frac{\Delta H}{T} \text{ at constant temperature and pressure} \quad (2)$$

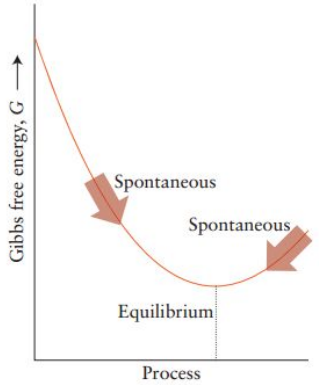


FIGURE 4J.2 At constant temperature and pressure, the direction of spontaneous change is toward lower Gibbs free energy. The horizontal axis represents the progress of the reaction or process. The equilibrium state of a system corresponds to the lowest point on the curve.

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Gibbs Free energy and non-expansion work

To find the relation between the Gibbs free energy and the maximum nonexpansion work, start with Eq. 3 for an infinitesimal change (denoted d) in G at constant temperature:

$$dG = dH - TdS \quad \text{at constant temperature}$$

Then use Eq. 1 of Topic 4C ($H = U + PV$) to express the infinitesimal change in enthalpy at constant pressure in terms of the change in internal energy and volume:

$$dH = dU + PdV \quad \text{at constant pressure}$$

dU+PdV

Next, substitute this expression into the preceding one:

$$dG = dU + PdV - TdS \quad \text{at constant temperature and pressure}$$

Now use Eq. 1 of Topic 4B for an infinitesimal change in internal energy ($dU = dw + dq$) and obtain

$$dG = dw + dq + PdV - TdS \quad \text{at constant temperature and pressure}$$

In order for a process to do the maximum possible work, it must occur reversibly. For a reversible change this equation becomes,

$$dG = dw_{rev} + dq_{rev} + PdV - TdS \quad \text{at constant temperature and pressure}$$

Now use the infinitesimal version of Eq. 1 of Topic 4F ($dS = dq_{rev}/T$) to replace dq_{rev} by TdS and cancel the two TdS terms:

$$\begin{aligned} dG &= dw_{rev} + TdS + PdV - TdS \\ &= dw_{rev} + PdV \quad \text{at constant temperature and pressure} \end{aligned}$$

At this point, note that the system may do both expansion work and nonexpansion work:

$$dw_{rev} = dw_{rev,\,e} + dw_{rev,\,expansion}$$

Reversible expansion work (achieved by matching the external to the internal pressure) is given by the infinitesimal version of Eq. 3 of Topic 4A ($w_{expansion} = -P_{ex}\Delta V$, which becomes $dw_{expansion} = -P_{ex}dV$), and setting the external pressure equal to the pressure of the gas in the system at each stage of the expansion,

$$dw_{rev,\,expansion} = -PdV$$

Reversible expansion work (achieved by matching the external to the internal pressure) is given by the infinitesimal version of Eq. 3 of Topic 4A ($w_{expansion} = -P_{ex}\Delta V$, which becomes $dw_{expansion} = -P_{ex}dV$), and setting the external pressure equal to the pressure of the gas in the system at each stage of the expansion,

$$dw_{rev,\,expansion} = -PdV$$

It follows that

$$dw_{rev} = dw_{rev,\,e} - PdV$$

When this expression is substituted into the last line of the expression for dG ($dG = dw_{rev} + PdV$), the PdV terms cancel:

$$dG = dw_{rev,\,e} + PdV - PdV$$

leaving

$$dG = dw_{rev,\,e} \quad \text{at constant temperature and pressure}$$

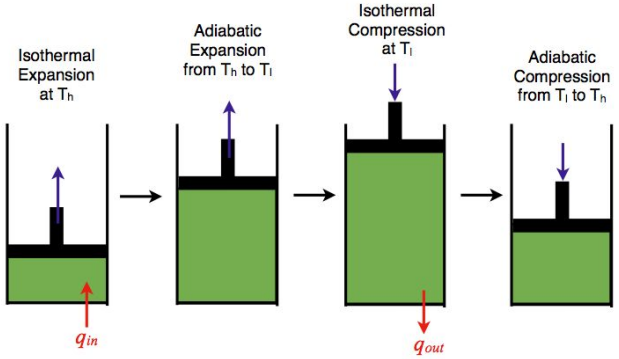
Since $dw_{rev,\,e}$ is the *maximum* nonexpansion work that the system can do (because it is achieved reversibly), the final result is

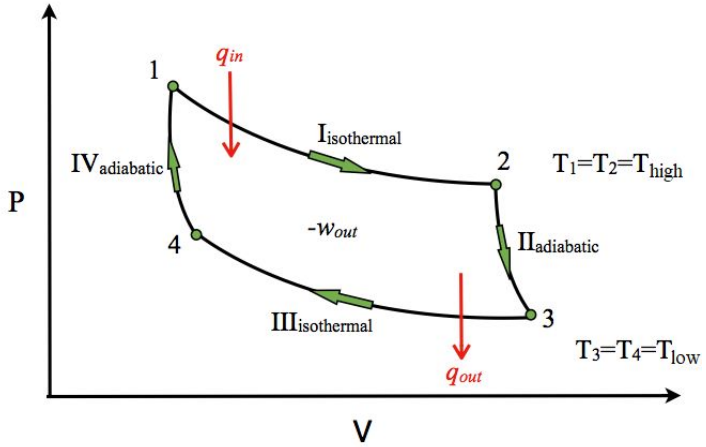
$$dG = dw_{e,\,max} \quad \text{at constant temperature and pressure}$$

ΔG for a process is equal to the maximum non-expansion work that the system can do at constant T and P.

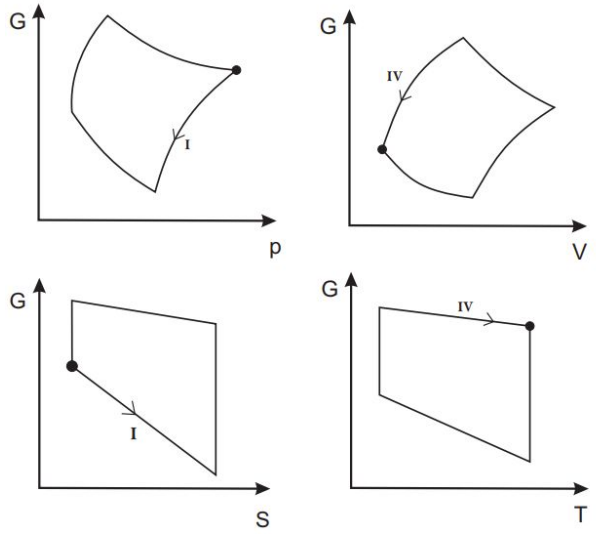
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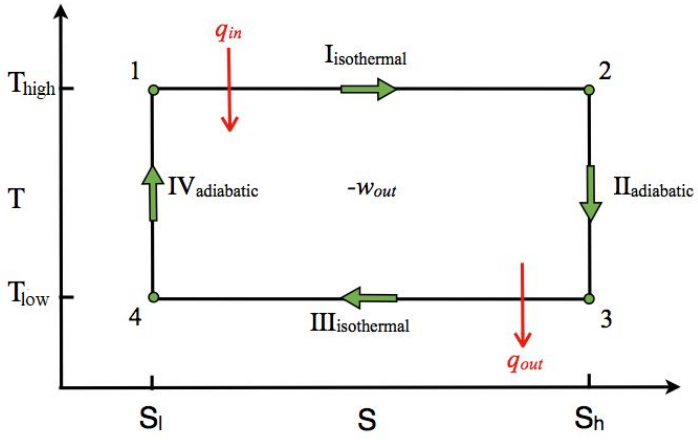
What about Carnot cycle?





$\Delta G=0$ for Carnot cycle





https://www.ias.ac.in/article/fulltext/reso/030/07/0957-0971

Thermodynamic identity for dG

$$G = U + PV - TS$$

$$dG = dU + PdV + VdP - TdS - SdT$$

$$Substituting\ dU = TdS - PdV + \mu dN$$

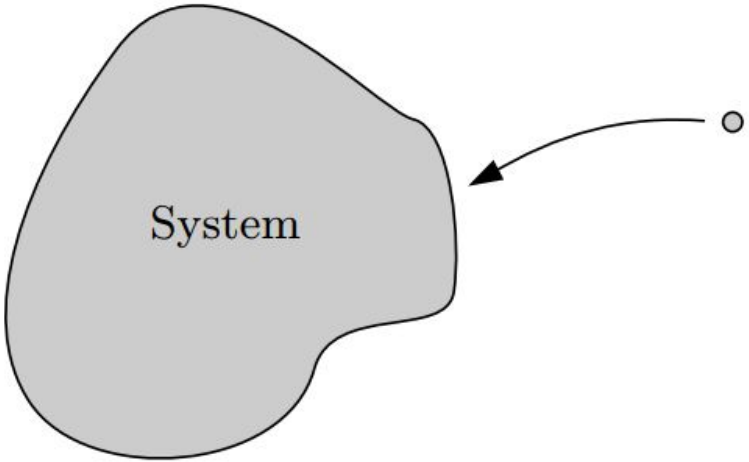
$$dG = TdS - PdV + \mu dN + PdV + VdP - TdS - SdT$$

$$dG = -SdT + VdP + \mu dN$$

$$S = -\left(\frac{\partial G}{\partial T}\right)_{P,N}, \qquad V = \left(\frac{\partial G}{\partial P}\right)_{T,N}, \qquad \boxed{\mu = \left(\frac{\partial G}{\partial N}\right)_{T,P}}$$

Chemical Potential

$$\mu = \left(\frac{\partial G}{\partial N}\right)_{T,P}$$

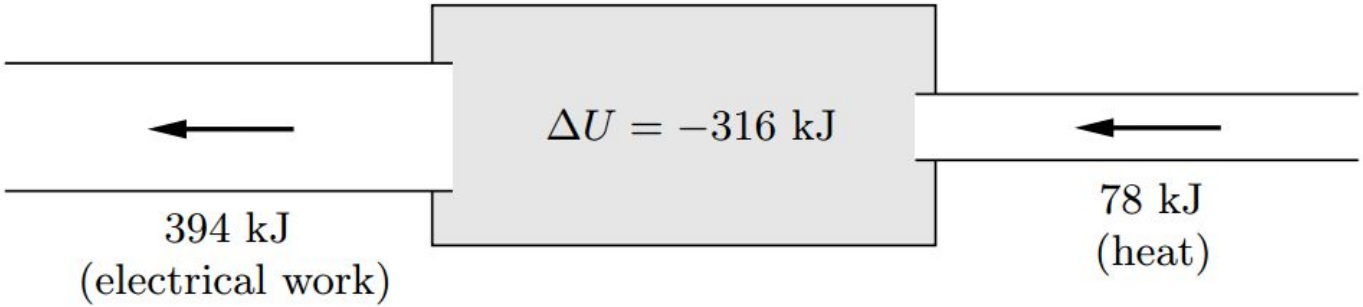


- When you add a particle to a system, holding the temperature and pressure fixed, the system's Gibbs free energy increases by μ
- Each additional particle must add exactly the same amount to G
- For a pure system with only one type of particle: **μ is just the Gibbs free energy per particle**

  μ = G/N

- Particles tend to flow from the system with higher μ into the system with lower μ

Case 1: Electrical engineer meets G



$$\begin{aligned}
\text{in solution:} \quad & 2\mathrm{SO_4^{2-}} + 2\mathrm{H^+} \longrightarrow 2\mathrm{HSO_4^-}; \\
\text{at } - \text{ electrode:} \quad & \mathrm{Pb} + \mathrm{HSO_4^-} \longrightarrow \mathrm{PbSO_4} + \mathrm{H^+} + 2\mathrm{e^-}; \\
\text{at } + \text{ electrode:} \quad & \mathrm{PbO_2} + \mathrm{HSO_4^-} + 3\mathrm{H^+} + 2\mathrm{e^-} \longrightarrow \mathrm{PbSO_4} + 2\mathrm{H_2O}
\end{aligned}$$

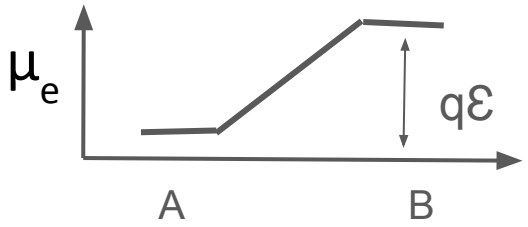
The electrical work produced *per electron* is

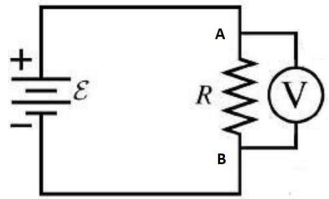
$$\frac{394 \text{ kJ}}{2 \cdot 6.02 \times 10^{23}} = 3.27 \times 10^{-19} \text{ J} = 2.04 \text{ eV}$$

A car battery contains six lead-acid cells, giving a total of about 12 V

Case 2: Electrical engineer meets μ

- We use chemical potential in Quantum statistics in calculating the probability of an electron occupying any energy level
- We have ohmic contacts for all electronic devices which can be used to apply voltage/current bias at that terminal.
- Chemical potential of electrons at that contact is raised (lowered) by the negative (positive) applied bias **(More in EE3001 (SSD))**
- A voltmeter connected between two terminals of an electronic component measures the volt eqvt of the difference between chemical potential of the two terminals



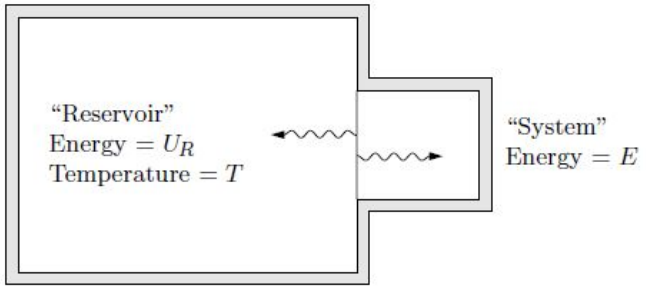


Canonical ensemble

- Consider a “system” in thermal contact with a much larger “reservoir” at some well-defined temperature.
- Reservoir + system can be treated like an isolated system
- $U_R + E = E^0 \Rightarrow U_R = E^0 - E$
- $P_s(E) = C\Omega_s(E) = P_R(E^0 - E) = C'\Omega_R(E^0 - E)$, where C & C’ are constants indpt of microstate
- If $E \ll E^0$, taking a Taylor expansion of $\ln \Omega_R(E^0 - E)$

  $$\ln \Omega_R(E^0 - E) = \ln \Omega_R(E^0) - E\left(\partial \ln \Omega_R / \partial U_R\right)_{E0} + \ldots.$$

- $S = k \ln \Omega$ and using the result, $\partial \ln \Omega / \partial U = 1/kT = \beta$
- $\Omega_R(E^0 - E) = \Omega_R(E^0)\, e^{-\beta E}$

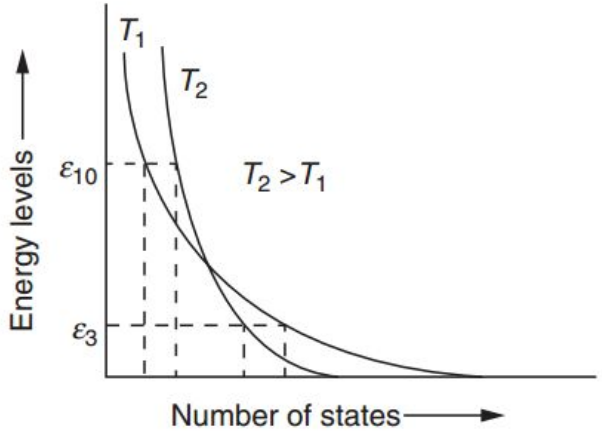


Boltzmann factor & Partition function

- $\Omega_R(E^0 - E) = \Omega_R(E^0)\, e^{-\beta E}$
- $\Omega_s(E) = (C'/C)\, \Omega_R(E^0)\, e^{-\beta E}$ => $P(E) = (1/Z) e^{-\beta E}$, where Z is constant which can be found using normalization
- Boltzmann factor = $e^{-\beta E}$
- Probability of finding the system in state s,

$$\mathcal{P}(s) = \frac{1}{Z} e^{-E(s)/kT}$$

where Z, which is called as the partition function



$$Z = \sum_s e^{-E(s)/kT}$$

- Average value of energy

$$\overline{E} = \frac{\sum_s E(s)N(s)}{N} = \sum_s E(s)\frac{N(s)}{N} = \sum_s E(s)\mathcal{P}(s) = \frac{1}{Z}\sum_s E(s)\, e^{-\beta E(s)}$$

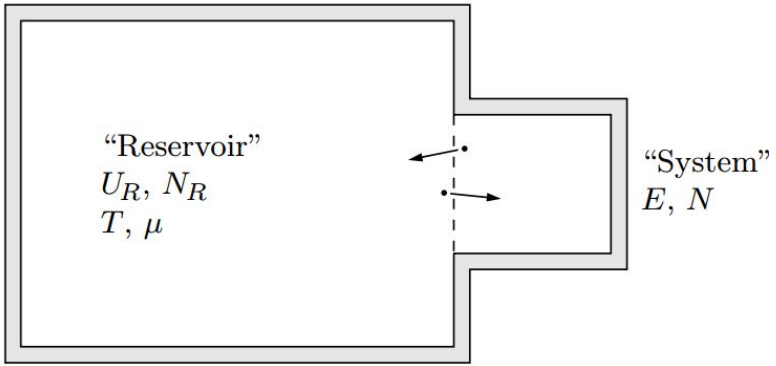
Grand canonical ensemble and Gibbs Factor

- A system in thermal and diffusive contact with a much larger reservoir, whose temperature and chemical potential are effectively constant.

- Gibbs factor = $e^{-[E(s)-\mu N(s)]/kT}$

- Grand partition function,

$$\mathcal{Z} = \sum_s e^{-[E(s)-\mu N(s)]/kT}$$



- Probability of finding the system in state s,

$$\mathcal{P}(s) = \frac{1}{\mathcal{Z}} e^{-[E(s)-\mu N(s)]/kT}$$

- **We can compute all the thermodynamic properties of a system, once we know its partition function**

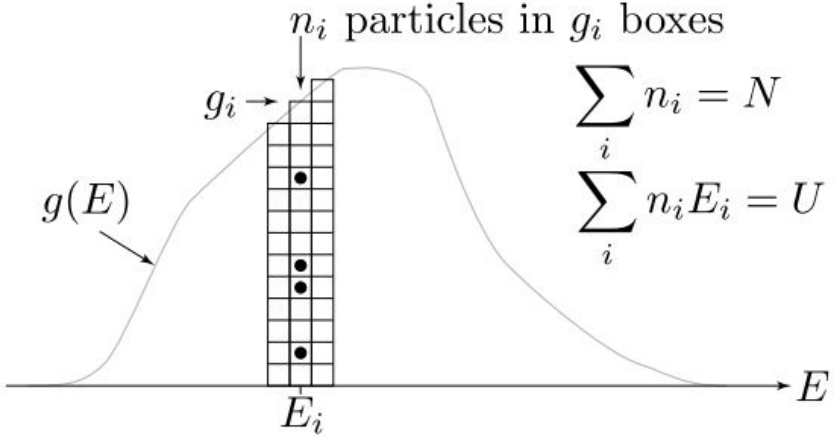
Classical and quantum mechanical particles

- In classical physics, we say that we can keep track of each specific particle by simply keeping track of its trajectory.
- As Quantum particles approach closer to each other, there is an increasing probability that they could interchange positions without our knowledge. In other words, identical particles could swap positions at any time without our knowledge, and hence these identical particles cannot be distinguished from each other.
- If the two particles are indistinguishable, the wave function must satisfy further requirements depending on whether they are Fermions or Bosons
- One consequence of this for Fermions is Pauli exclusion principle.

Classical particles following Max-Boltz statistics

- How many ways you can put two particles in three states?
- Distinguishable particles
- Any number of particles can be in any one state

1	2	3
AB	. . .	. . .
. . .	AB	. . .
. . .	. . .	AB
A	B	. . .
B	A	. . .
A	. . .	B
B	. . .	A
. . .	A	B
. . .	B	A



$$\sum_i n_i = N$$

$$\sum_i n_i E_i = U$$

$$W_B = N! \prod_{i=0}^{\infty} \frac{(g_i)^{N_i}}{N_i!}$$

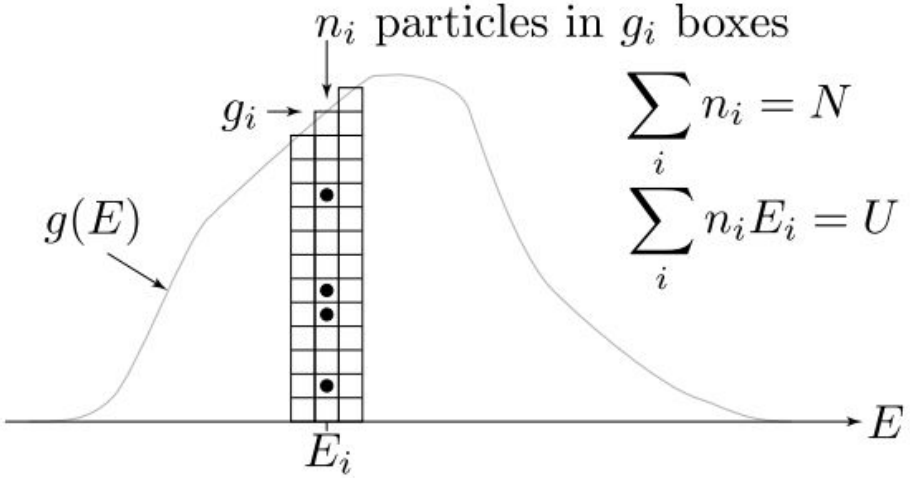
g(E) refers to density of states (DOS) to occupy which will be covered later

g_i=3, N=N_i=2 and W_{MB}=9

Quantum particles following Bose-Einstein statistics

- How many ways you can put two particles in three states?
- Indistinguishable particles
- Any number of particles can be in any one state

1	2	3
AA	. . .	. . .
. . .	AA	. . .
. . .	. . .	AA
A	A	. . .
A	. . .	A
. . .	A	A



$$\sum_i n_i = N$$

$$\sum_i n_i E_i = U$$

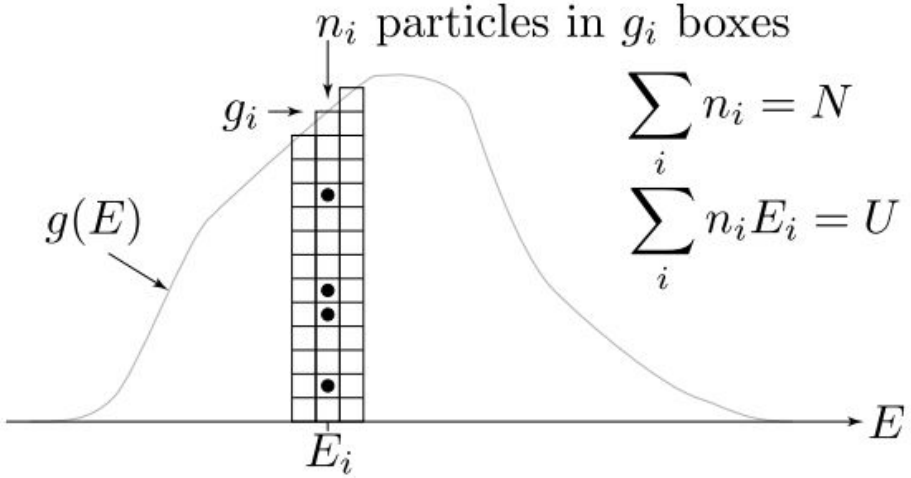
$$\Omega = \prod_i \frac{(g_i + n_i - 1)!}{n_i!(g_i - 1)!}$$

g_i=3, n_i=2 and W_{BE}=6

Quantum particles following Fermi-Dirac statistics

- How many ways you can put two particles in three states?
- Indistinguishable particles
- Pauli's exclusion principle - No more than one particle can be in any one state

1	2	3
A	A	. . .
A	. . .	A
. . .	A	A



$$\sum_i n_i = N$$

$$\sum_i n_i E_i = U$$

$$\Omega = \prod_i \frac{g_i!}{n_i!(g_i - n_i)!}$$

g_i=3, n_i=2 and W_{FD}=3

Non-interacting Fermions

Consider a system of non-interacting fermions. The fermions can occupy microscopic quantum states with energies ε_i. Since the fermions do not interact, the energy of one quantum state does not depend on the occupation of any of the other quantum states. Because of the Pauli principle, each (spin specific) microscopic state only be occupied by either zero or one fermion. A macrostate q of this system consists of N_q fermions. The total energy of this macrostate is E_q.

$$N_q = \sum_i n_{qi} \qquad E_q = \sum_i n_{qi}\varepsilon_i \qquad n_{qi} \in 0,1$$

Here n_{qi} are occupation numbers that specify if microstate i is occupied in macrostate q.

To calculate the thermodynamic properties of a system of non-interacting fermions, the grand canonical partition function Z_{gr} is constructed.

$$Z_{gr} = \sum_q \exp\left(\frac{\mu}{k_B T}\right)^{N_q} \exp\left(-\frac{E_q}{k_B T}\right) = \sum_q \exp\left(-\frac{E_q - \mu N_q}{k_B T}\right)$$

The sum q runs over all of the possible macroscopic states, μ is the chemical potential, k_B is Boltzmann's constant, and T is the absolute temperature. The energy and particle number of the macrostates can be expressed in terms of the microscopic states.

$$Z_{gr} = \sum_q \exp\left(-\frac{\sum_i n_{qi}(\varepsilon_i - \mu)}{k_B T}\right) = \sum_q \prod_i \exp\left(-\frac{n_{qi}(\varepsilon_i - \mu)}{k_B T}\right)$$

Non-interacting Fermions

The sum over all possible macrostates can also be written as the sum over all possible microstates.

$$Z_{gr} = \sum_{n_0=0}^{1} \sum_{n_1=0}^{1} \cdots \sum_{n_{q\max}=0}^{1} \prod_i \exp\left(-\frac{n_{qi}\left(\varepsilon_i - \mu\right)}{k_B T} \right)$$

The microstate occupation numbers are either 0 or 1 so this can be written as,

$$Z_{gr} = \prod_i \left(\exp\left(-\frac{\left(\varepsilon_i - \mu\right)}{k_B T} \right) + 1 \right).$$

$$\mathcal{Z} = \prod_{\text{single particle states } i} \mathcal{Z}_i$$

The full grand-canonical partition function is then

$$\mathcal{Z} = \prod_i \mathcal{Z}_i = \prod_i \left[1 + e^{-\beta(\varepsilon_i - \mu)}\right]$$

The grand canonical ensemble makes it very easy to write an expression for the expected occupation number of state i:

$$\langle n_i \rangle = \frac{1}{\mathcal{Z}_i} \sum_{n_i} n_i e^{-n_i \beta(\varepsilon_i - \mu)} = \frac{1}{\beta} \frac{\partial \ln \mathcal{Z}_i}{\partial \mu} \tag{20}$$

We can also express this as

$$\langle n_i \rangle = -\frac{\partial \Phi_i}{\partial \mu} \tag{21}$$

where

$$\Phi_i = -\frac{1}{\beta} \ln \mathcal{Z}_i \tag{22}$$

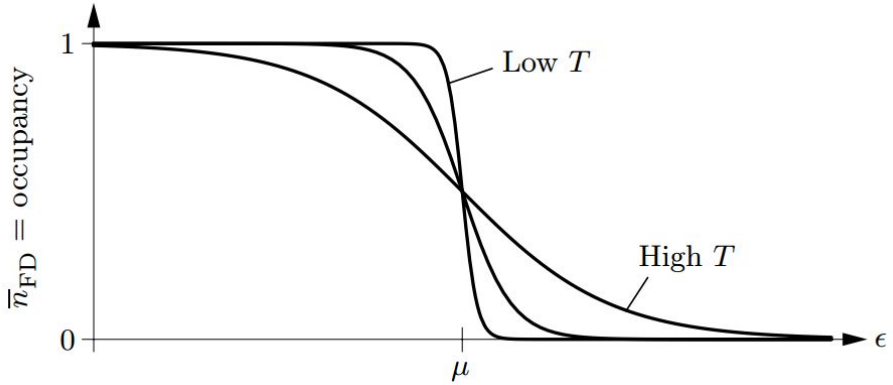
is the grand free energy for state i.

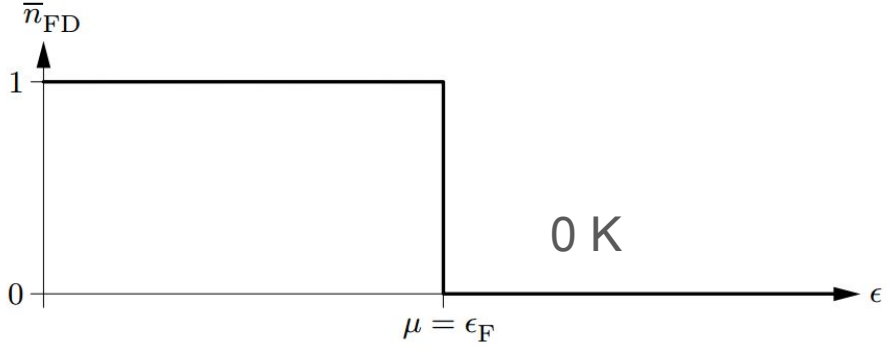
So that the occupation number for state i is

$$\langle n_i \rangle = -\frac{\partial \Phi_i}{\partial \mu} = \frac{1}{e^{\beta(\varepsilon_i - \mu)} + 1} \equiv f(\varepsilon)$$

Probability of an electron to occupy the energy level, ε_i

Fermi-Dirac Distribution





At T = 0, the Fermi-Dirac distribution equals 1 for all states with $\epsilon < \mu$ and equals 0 for all states with $\epsilon > \mu$. $\mu(T=0)$ is also called the Fermi Energy.

Solid State Physics and Electrical Engg text books continue to use Fermi energy or Fermi level (E_F) to represent μ even at T>0

For T>0, the FD distribution goes to zero when $\epsilon >> \mu$, and goes to 1 when $\epsilon << \mu$.

Non-interacting Bosons

$$\mathcal{Z} = \sum_{N,E} e^{-\beta(E-N\mu)}$$

Since $\sum n_i = N$ and $\sum n_i \varepsilon_i = E$ the grand partition function can also be written as

$$\mathcal{Z} = \prod_{\mathrm{single particle states}\, i} \mathcal{Z}_i$$

For a Bose system, there can be any number $n_i = 0, 1, 2, \cdots$ of bosons in a given state i. The single-state grand-canonical partition function for the state i with energy ε_i, from Eq. (19), is then

$$\mathcal{Z}_i = \sum_{n=0}^{\infty} e^{-n\beta(\varepsilon_i-\mu)} \tag{23}$$

This is a simple geometric series: there can be 0 particles, 1 particles, 2 particles, etc in the state, and so the total amount of energy in the state is an integer multiple of ε_i. Performing the sum, we find

$$\mathcal{Z}_i = \sum_{n=0}^{\infty} e^{-n\beta(\varepsilon_i-\mu)} = \frac{1}{1-e^{-\beta(\varepsilon_i-\mu)}} \tag{24}$$

The full partition function from Eq. (18) is then

$$\mathcal{Z} = \prod_i \mathcal{Z}_i = \prod_i \frac{1}{1-e^{-\beta(\varepsilon_i-\mu)}} \tag{25}$$

Bose-Einstein Distribution

The grand free energy for each state is

$$\Phi_i = -\frac{1}{\beta}\ln \mathcal{Z}_i = \frac{1}{\beta}\ln(1 - e^{-\beta(\varepsilon_i - \mu)})$$

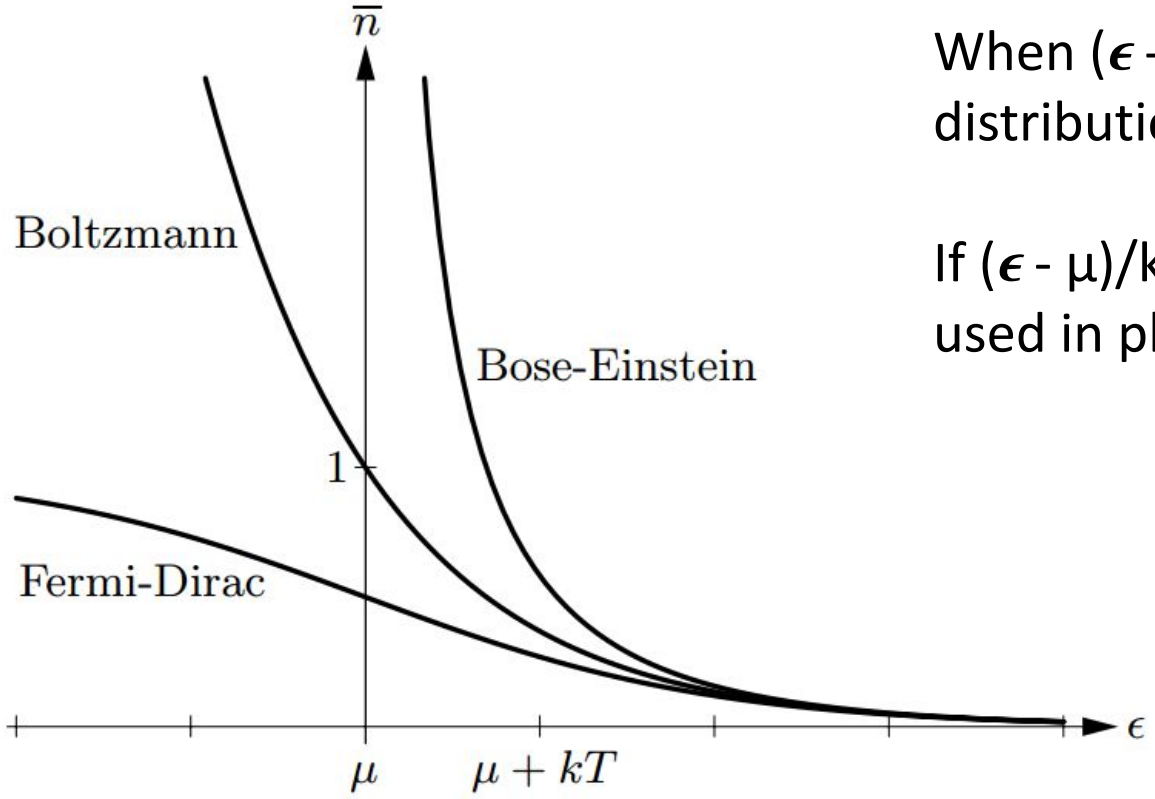
Then the occupation number for each state is

$$\langle n_i \rangle = -\frac{\partial \Phi_i}{\partial \mu} = \frac{1}{e^{\beta(\varepsilon_i - \mu)} - 1}$$

$$\overline{n}_{\mathrm{BE}} = \frac{1}{e^{(\epsilon - \mu)/kT} - 1}$$

μ must be always less than ϵ. BE distribution goes to zero when $\epsilon >> \mu$. It goes to infinity as ϵ approaches μ from above

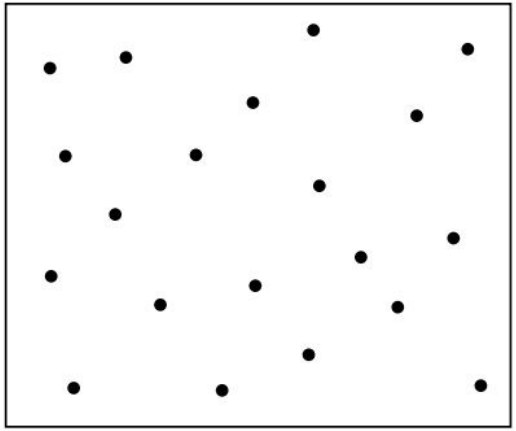
Approximating as Maxwell-Boltzmann Distribution



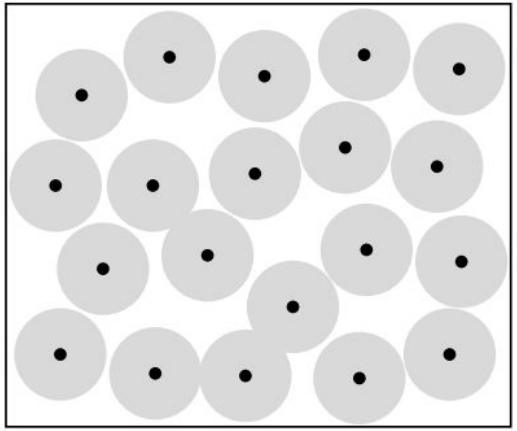
When $(\epsilon - \mu)/kT >> 1$, the three distributions become equal

If $(\epsilon - \mu)/kT > 3$, MB can be used in place of others

Approximating as Maxwell-Boltzmann Distribution



Normal gas, $V/N \gg v_Q$

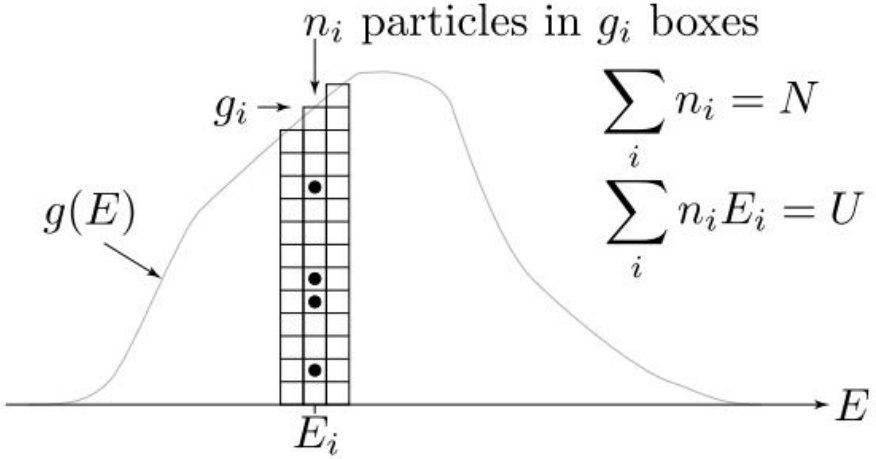


Quantum gas, $V/N \approx v_Q$

$$v_Q = \ell_Q^3 = \left(\frac{h}{\sqrt{2\pi mkT}}\right)^3$$

When the average distance between quantum gas particles is much greater than the average de Broglie wavelength, they can be treated like normal gas molecules. Is it valid for electrons in metals? Check

Are we ready to take a total count of electrons?

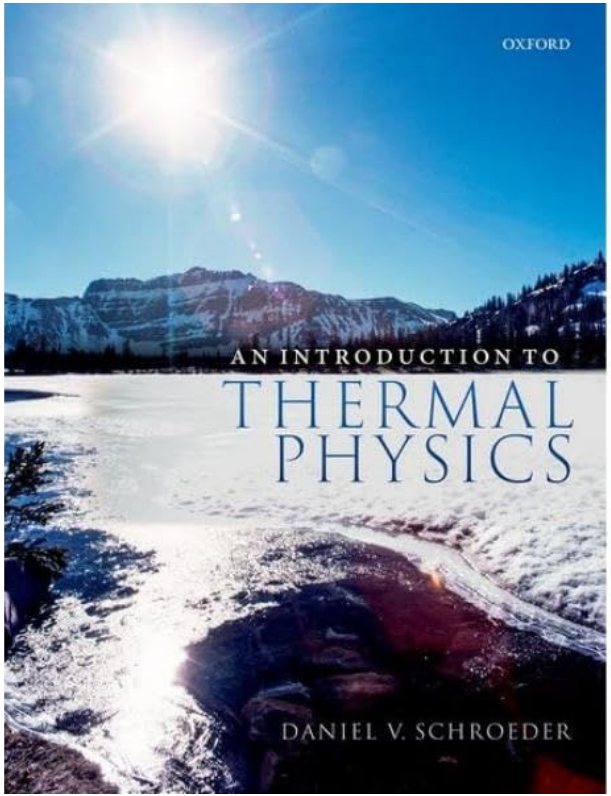




The number of single-particle states per unit energy, g(E) is called the density of states (DOS).

$$N = \int_0^\infty g(\epsilon)\,\overline{n}_{\mathrm{FD}}(\epsilon)\,d\epsilon$$

Reference


OXFORD
AN INTRODUCTION TO
THERMAL
PHYSICS
DANIEL V. SCHROEDER